

Peer Response

by Mansour Saeed Mubarak Hamdan Al Hamdani - Monday, 11 August 2025, 8:17 AM
Your association of ABS with decentralized company functioning is well-grounded, contrasting with centralized models and supported by Ding et al. (2023) and Sayyed-Alikhani et al. (2021). The emphasis on autonomy for real-time decision-making in supply chains is convincing, especially as it links to improved decision speed and responsiveness. Your recognition of ABS's potential for optimizing resource allocation; reinforced by Mueller-Bloch et al. (2024) in decentralized blockchain governance; broadens the discussion to governance and infrastructure design. Your reference to Aftabi et al. (2025) on mitigating threats in adaptive networks is valuable, as such systems can suffer cascading failures if coordination fails. Your caution on overdependence on agents is valid, as flawed decision rules or biased data can amplify systemic risks. Ethical adaptability could involve transparent decision protocols, continuous auditing, and hybrid human; AI oversight. Your suggestion on customer service automation adds another relevant application: ABS could enable personalized, real-time service delivery via adaptive virtual agents, while coordinating across numerous back-end systems, enhancing scalability and responsiveness in high-volume environments.

References

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Sayyed-Alikhani, M., Abbasi, M. & Sadeghi-Niaraki, A. (2021) 'A comprehensive review of multi-agent systems in supply chain management', *Journal of Industrial Information Integration*, 24, p. 100224.

Mueller-Bloch, C., Andersen, J.V., Spasovski, J. & Hahn, J. (2024) 'Understanding decentralization of decision-making power in proof-of-stake blockchains: An agent-based simulation approach', *European Journal of Information Systems*, 33(3), pp. 267–286.

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by Mansour Saeed Mubarak Hamdan Al Hamdani - Wednesday, 3 September 2025, 5:54 PM
Your discussion on Agent Communication Languages (ACLs), particularly KQML, effectively highlights their role in fostering cooperation and coordination in multi-agent systems (MAS). I agree that ACLs' focus on knowledge and intention transfer, as noted by Ma et al. (2024), enables them to thrive in open, heterogeneous environments, making them ideal for applications like e-commerce and supply-chain management (Allouch et al., 2021). The scalability you mention, supported by Cao et al. (2024), is a significant advantage, allowing seamless integration of new agents.

Your analysis of ACLs' drawbacks—requiring sophisticated reasoning and incurring processing costs (Kuhail et al., 2023)—is insightful. The inefficiency compared to method invocation in Python or Java, which excels in tightly coupled systems, underscores a clear trade-off (Liu et al., 2025). Your point about ACLs' limited use in time-sensitive fields like robotics, favoring simpler industry solutions, aligns with current trends.

To build on this, it could be worth exploring how advancements in AI might reduce ACLs' computational overhead or enhance their real-time applicability. Your balanced view on flexibility versus speed is compelling—great work, and I'm eager to hear more on future ACL innovations.

References

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